
MECHANISM-DRIVEN OPTIMIZATION OF YOLO DETECTORS ON PLANTDOC

PARADIGMS, ATTENTION, AND TRAINING STRATEGIES

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ABSTRACT

Automatic plant disease detection from leaf images faces significant challenges, including minute lesion sizes, complex background clutter, and severe class imbalance, particularly when annotated data is scarce. In this study, we present a mechanism-driven analysis to identify the performance bottlenecks of YOLO-style detectors on the PlantDoc dataset. Moving beyond simple hyperparameter tuning, we design controlled experiments across three strategic dimensions: (i) Detection Paradigms, comparing anchor-based (YOLOv5s/m) against anchor-free (YOLOX-s) architectures; (ii) Architectural Enhancement, integrating a lightweight Convolutional Block Attention Module (CBAM) into YOLOv5s; and (iii) Optimization Strategies, replacing standard NMS with Soft-NMS and adopting Focal Loss to mitigate occlusion and class imbalance. All models were trained under a strictly controlled environment with identical splits, fixed seeds, and 640×640 resolution. Our results indicate that model capacity is a primary performance driver, with YOLOv5m outperforming YOLOv5s by approximately X.X% mAP@0.5. The anchor-free YOLOX-s achieves competitive performance with a distinct precision-recall trade-off. Notably, incorporating attention (CBAM) into the lightweight YOLOv5s yields a gain of X.X% mAP@0.5, effectively narrowing the performance gap to the larger YOLOv5m with negligible computational cost. Conversely, optimization strategies like Soft-NMS and Focal Loss primarily re-balanced Precision and Recall rather than improving aggregate mAP, suggesting that intrinsic data scarcity and inter-class visual overlap remain the dominant bottlenecks on PlantDoc.

Keywords Plant Disease Detection · Object Detection · YOLO · Attention Mechanism · Class Imbalance · Small Object Detection

1 Introduction

Plant diseases pose a significant economic and yield threat to agriculture worldwide [1]. Timely and accurate diagnosis is critical for effective disease management. The advent of deep learning, particularly Convolutional Neural Networks (CNNs), marked a paradigm shift from labor-intensive manual inspection, demonstrating unprecedented success in automated visual recognition for plant disease detection in controlled environments [2]. The central motivation of this study is to move beyond these initial laboratory successes and conduct a rigorous technical analysis to bridge the performance gap between ideal data environments and the variable conditions required for robust, large-scale automated detection systems.

However, deploying fast object detectors in chaotic agricultural settings remains critically challenging. Models trained on clean data degrade significantly when faced with complex, “in-the-wild” datasets like PlantDoc [2], which captures

realistic difficulties. These difficulties, which constitute the specific technical challenges addressed in this study, include the issue that early-stage lesions are often extremely small and occupy only a tiny fraction of the image. This results in a severe small object detection problem, where discriminative features are easily lost in deeper convolutional layers. Second, agricultural images frequently contain dense foliage where leaves overlap, causing occlusion and missed detections. Third, disease occurrence follows a long-tail distribution, resulting in severe class imbalance, where majority classes dominate the loss function during training.

While state-of-the-art one-stage detectors like YOLO (You Only Look Once) are popular for their speed, applying them naively to complex, unbalanced data like PlantDoc yields suboptimal results due to architectural bottlenecks and inherent algorithmic limitations [3]. Our primary objective is to move beyond superficial hyperparameter tuning to conduct a rigorous, mechanism-driven analysis. We aim to systematically isolate and measure the impact of fundamental architectural and algorithmic changes specifically designed to overcome the small object, occlusion, and class imbalance problems endemic to object detection on the PlantDoc dataset [2].

In this work, we move beyond simple parameter tuning and propose a mechanism-driven analysis to systematically optimize YOLO-style detectors for plant disease detection on the PlantDoc dataset.

The systematic experimental framework addressing these deficiencies involves investigating different detection paradigms (Anchor-Based vs. Anchor-Free), incorporating architectural enhancements (CBAM), and evaluating advanced optimization strategies (Focal Loss, Soft-NMS). The key outcomes of this comprehensive study are summarized below:

1. We demonstrate that the Anchor-Free paradigm (YOLOX) offers inherent advantages in precision-recall trade-offs for irregular biological targets compared to traditional Anchor-Based methods.
2. We show that architectural modifications, specifically the insertion of CBAM, provide a cost-effective performance boost for lightweight models, partially bridging the gap to larger models.
3. We provide a critical analysis of long-tail optimization strategies, revealing that while methods like Focal Loss alter the precision-recall balance, they are not a silver bullet for data scarcity in agricultural datasets.
4. We provide a valuable resource to the community by creating and publicly releasing a standardized YOLOX-compatible annotation format for the PlantDoc dataset (to be hosted on Kaggle), enabling future Anchor-Free research on this challenging resource.

2 Related Work

2.1 Plant Disease Detection and the PlantDoc Dataset

Early work on plant disease detection relied on handcrafted features and traditional classifiers, and was typically evaluated on controlled datasets with clean backgrounds and limited variability. With the advent of deep learning, CNN-based models have become the dominant approach for plant disease classification. Mohanty et al. [1] trained deep convolutional networks on the PlantVillage dataset, achieving over 99% accuracy under controlled imaging conditions. While these results demonstrate the feasibility of image diagnosis, they also highlight a substantial gap to real field conditions.

To address this gap, Singh et al. [2] introduced the PlantDoc dataset, which consists of scraped leaf images of multiple crops with diverse backgrounds, lighting conditions and viewpoints. Their study showed that CNN classifiers on PlantDoc improves performance compared to models trained solely on lab datasets, and emphasised the need for uncontrolled data in plant disease detection. More recent work has further explored object detection on leaf images, but most studies focus on improving overall accuracy rather than understanding failure modes on small lesions and long-tailed classes.

In this project, we adopt the object detection variant of PlantDoc, where diseased and healthy leaves are annotated with bounding boxes. Rather than proposing yet another detector, we use PlantDoc as a testbed for a mechanistic analysis of YOLO detectors under realistic conditions of limited data, small lesion size, and class imbalance.

2.2 YOLO Detectors and Detection Paradigms

Single-stage detectors in the YOLO family [4] have become a standard choice for real-time object detection due to their favourable accuracy and speed trade-off. Successive versions have improved backbone and neck design, training strategies and implementation details, leading to strong performance on benchmarks such as MS COCO. YOLOv5 adopts CSP backbones and a PANet neck, and has been widely used as an off the shelf detector in many applied

domains. YOLOX [3] further pushes the design by switching to an anchor free detection head, introducing a decoupled classification and regression head and employing the SimOTA dynamic label assignment strategy, which achieves state of the art results across a range of model sizes.

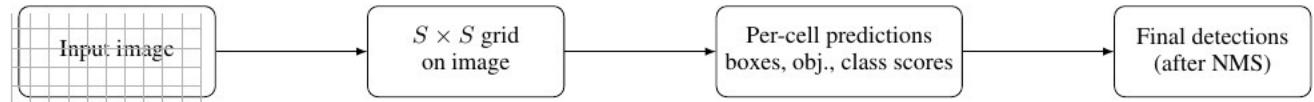


Figure 1: Simplified YOLO-style detection pipeline: the image is divided into an $S \times S$ grid and each cell predicts bounding boxes with confidences and class scores, followed by NMS.

In the context of agriculture, a number of works employ YOLO detectors for plant disease detection and pest monitoring, mainly reporting mAP improvements over earlier two stage detectors or pure classification pipelines. However, these studies often treat the detector as a black box and rarely disentangle how architectural choices, such as anchor based vs. anchor free design or model capacity, affect small lesion recall and performance on long tail classes.

In contrast, we explicitly treat YOLOv5s, YOLOv5m and YOLOXs as representative design points along two axes: capacity (small vs. medium model) and detection paradigm (anchor based vs. anchor free with dynamic label assignment). Our work thus complements existing results by using these detectors to probe the architectural and assignment mechanisms that matter most on PlantDoc, rather than simply reporting another set of mAP numbers.

2.3 Attention Mechanisms, Soft-NMS and Focal Loss

Attention mechanisms have been widely adopted to enhance feature representations in CNNs. The Convolutional Block Attention Module (CBAM) [5] combines channel and spatial attention in a lightweight block that can be inserted into existing architectures with negligible overhead. CBAM and related modules have been shown to improve both image classification and object detection by reweighting informative channels and regions while suppressing background clutter. Recent agricultural vision work has integrated CBAM into YOLO detectors to better capture fine grained lesion patterns and complex backgrounds.

Beyond architectural enhancements, several training and inference mechanisms have been proposed to address typical failure modes of dense detectors. Soft-NMS [6] replaces the hard removal in standard non-maximum suppression with a continuous decay of detection scores as a function of IoU, improving average precision in crowded scenes by reducing missed true positives due to aggressive suppression. Focal Loss [7] reshapes the standard cross-entropy/BCE loss with a focusing term to down weight easy examples and emphasise hard, misclassified ones, and has become a standard tool for addressing class imbalance in single stage detectors such as RetinaNet.

Most prior work evaluates CBAM, Soft-NMS and Focal Loss on large scale benchmarks with abundant training data. In this project, we adopt these mechanisms as components for YOLOv5: we insert CBAM into the lightweight YOLOv5s backbone, and apply Soft-NMS and Focal Loss (with fixed focusing parameter) to the higher capacity YOLOv5m model. Our goal is not to introduce new variants, but to systematically assess how much these widely used mechanisms actually help or fail to help on a small, imbalanced dataset like PlantDoc, in line with our mechanism driven objectives.

3 Methods

This section details the design and configuration of the mechanism-driven experimental framework used to systematically analyze the performance bottlenecks of YOLO-style detectors on the PlantDoc dataset.

3.1 Dataset and Experimental Protocol

3.1.1 PlantDoc Dataset

The PlantDoc dataset [2] was selected as the testing environment due to its explicit focus on capturing realistic, “in-the-wild” conditions, which introduce environmental noise and complexity absent in traditional laboratory datasets such as PlantVillage [11]. The YOLO-formatted PlantDoc variant used in this work comprises approximately 2,600 images annotated with 29 leaf-level classes, each corresponding to a specific combination of plant species and symptom (e.g.,

Apple Scab Leaf, Tomato leaf bacterial spot, Potato leaf late blight). These annotations reflect the following real-world challenges pertinent to this study:

- **Small Object Detection:** Early-stage or distant lesions frequently result in objects occupying only a small number of pixels, which makes it difficult for deep convolutional detectors to preserve discriminative features in later layers [3].
- **Occlusion and Background Clutter:** The images often contain overlapping leaves and complex natural backgrounds, so target lesions are partially occluded or visually blended into the surroundings, degrading the generalization capacity of standard object detectors [6].
- **Class Imbalance:** The non-uniform occurrence of diseases in field environments induces a long-tailed label distribution, biasing conventional training objectives toward majority classes and hindering robust learning of rare categories [7].

These three factors collectively make PlantDoc a challenging benchmark for object detection. In this work, they directly motivate the mechanism-driven interventions studied in later sections, including anchor-free detection, attention-based architectural enhancements, and long-tail optimization strategies such as focal loss and Soft-NMS.

Note on Dataset Format for YOLOX. Given that the original PlantDoc annotations lack a publicly available, standardized format compatible with the anchor-free detection paradigm (YOLOX-s) [3], a custom conversion script was developed in this study. This script transforms the original bounding box annotations into the required format for YOLOX’s specific label assignment mechanism. This transformed dataset version will be uploaded to Kaggle to support future community research utilizing anchor-free architectures on PlantDoc.

The Kaggle PlantDoc object detection dataset provides a predefined train/validation split with 2,335 training images and 236 validation images (approximately 90.8% / 9.2%). We use the existing data partitions and choose not to reserve a separate test set. This retains more labeled samples for training, which is important for addressing the dataset’s small size and imbalanced distribution. All model comparisons are therefore reported solely on the designated validation split.

3.1.2 Controlled Training Environment

To ensure result validity and reproducibility, all models were trained within a strictly controlled experimental environment:

- **Input Resolution:** All input images were uniformly resized to 640×640 pixels, adhering to standard configurations used in recent YOLO-style detectors [3].
- **Reproducibility:** A fixed random seed was utilized for network initialization and data splitting across all experiments.
- **Hyperparameters:** Core training hyperparameters, including learning rate, optimizer, batch size, and weight decay, were kept identical across all compared models unless the hyperparameter itself was the mechanism being tested.

All YOLOv5-based models were trained for 50 epochs with an input resolution of 640×640 pixels and a batch size of 16, starting from COCO-pretrained weights (`yolo5s.pt` / `yolo5m.pt`). Optimization followed the default YOLO-style configuration, using stochastic gradient descent (SGD) with momentum 0.937 and weight decay $\approx 5 \times 10^{-4}$, combined with gradient accumulation to match a nominal batch size of 64. A warm-up phase of the first 3 epochs was applied to both the learning rate and momentum, after which a nearly constant learning rate of 0.01 was maintained via the linear learning-rate scheduler (with $lr_0 = lrf = 0.01$).

Data augmentation followed the standard YOLOv5 pipeline, including HSV-based color jitter, random translation (up to 10%), multi-scale resizing (scale factor up to 0.5), and horizontal flipping with probability 0.5. Mosaic augmentation was enabled throughout training, while more aggressive strategies such as mixup and copy-paste were disabled. An exponential moving average (EMA) of model weights was maintained during training, and the EMA weights were used for evaluation.

3.2 Baseline Models and Training Configuration

The YOLO family of models was chosen for its real-time performance capabilities, making it a relevant architecture for practical object detection in agriculture.

Baseline Selection. YOLOv5s (Small) was established as the primary baseline for modifications and enhancements. YOLOv5m (Medium) was included to quantify the impact of increasing baseline model capacity on performance against the inherent complexity of the PlantDoc dataset [2].

General Training. All models utilized transfer learning, initialized with weights pre-trained on the COCO dataset, and were fine-tuned using the PlantDoc training set.

3.3 Mechanism-Driven Experimental Design

The controlled experimentation was structured across three strategic dimensions, each directly targeting a specific deficiency of “in-the-wild” detection on the PlantDoc dataset.

3.3.1 Detection Paradigms (Anchor-Based vs. Anchor-Free)

To determine the most suitable object localization mechanism for irregular biological targets, we investigated two competing paradigms:

- **Anchor-Based:** YOLOv5s and YOLOv5m utilize predefined anchor boxes to facilitate bounding-box regression.
- **Anchor-Free:** YOLOX-s was employed as a contemporary alternative. YOLOX eliminates the need for predefined anchors, using a coupled head and the SimOTA advanced label assignment strategy, and has been shown to outperform previous YOLO variants on several benchmarks [3]. This comparison aims to assess whether the inherent flexibility of the anchor-free paradigm offers advantages in detecting the irregular, varied shapes of plant lesions.

3.3.2 Architectural Enhancement (CBAM Integration)

To mitigate the small object detection challenge, a lightweight attention mechanism was introduced.

CBAM Implementation. The Convolutional Block Attention Module (CBAM), which consists of channel and spatial attention sub-modules, was strategically integrated into the backbone of the lightweight YOLOv5s architecture [5]. The CBAM insertion is designed to force the model to explicitly weigh and focus computational resources on the diminutive, discriminative lesion features while suppressing irrelevant background information. The resulting performance gain was evaluated against the increased computational cost (GFLOPs and number of parameters).

3.3.3 Optimization Strategies (Focal Loss and Soft-NMS)

To address the challenges of class imbalance and occlusion in dense prediction areas, two advanced algorithmic strategies were implemented. These strategies were specifically evaluated by integrating them into the higher-capacity baseline model, YOLOv5m, to diagnose their impact beyond model scaling.

Focal Loss. To counter the long-tail class imbalance, the standard cross-entropy loss was replaced with focal loss [7]. This function dynamically down-weights the contribution of easily classified examples (majority/background classes) during training, thereby prioritizing the learning signal from hard, rare, and minority class samples (hard-mining).

Soft-NMS. To improve recall in scenes with heavy occlusion, the traditional hard-threshold non-maximum suppression (NMS) was replaced with Soft-NMS [6]. Soft-NMS retains highly overlapping bounding boxes by smoothly reducing their confidence scores, preventing the premature suppression of potentially correct detections in crowded, occluded foliage.

3.4 Evaluation Metrics

Model performance was rigorously quantified using standard object detection metrics:

- **Primary Metric:** The aggregate performance measure is mAP@0.5 (mean average precision at IoU = 0.5).
- **Diagnostic Metrics:** Precision (P) and recall (R) were recorded to provide granular insights into model behavior. Specifically, the trade-off between precision and recall was analyzed to diagnose the effects of the optimization strategies (focal loss, Soft-NMS) on the model’s conservatism versus its ability to locate all targets.

4 Results

This section reports the empirical results of the three experimental axes described in Section 3: detection paradigms (anchor-based vs. anchor-free), architectural enhancement via CBAM, and long-tail optimization strategies based on focal loss and Soft-NMS.

As illustrated in Table II, classes with abundant, visually distinctive samples (e.g., Cherry, Blueberry, Raspberry) achieve AP above 0.8, whereas rare or subtle categories such as Soyabean and Squash Powdery mildew remain below 0.2 AP, and some extremely underrepresented classes (Tomato leaf) cannot be reliably evaluated at all.

4.1 Detection Paradigms: YOLOv5s, YOLOv5m, and YOLOX-s

Table II summarizes the overall performance of the three detector families on the PlantDoc validation set. Scaling the anchor-based baseline from YOLOv5s to YOLOv5m yields a moderate improvement in both mAP@0.5 (from 0.633 to 0.647) and mAP@0.5:0.95 (from 0.485 to 0.506), indicating that additional capacity is beneficial under the challenging, long-tailed conditions of PlantDoc.

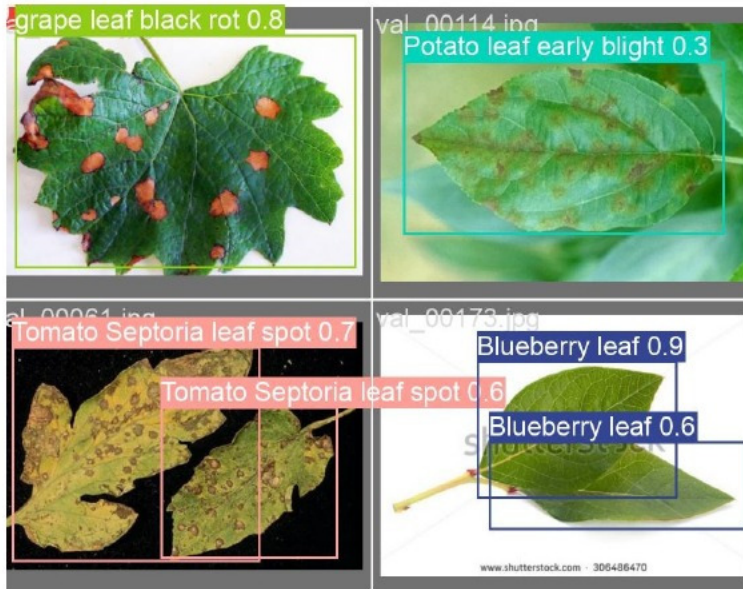


Figure 2: Qualitative YOLOX-s detections on PlantDoc validation images, illustrating correct localization under clutter and varying lesion sizes but low confidence on small or subtle symptoms.

The anchor-free YOLOX-s achieves the highest mAP@0.5 (0.661), slightly surpassing YOLOv5m, while attaining a comparable mAP@0.5:0.95 (0.504). A scale-sensitive breakdown of YOLOX-s reveals that performance is strongest on large lesions ($AP_{\text{large}} = 0.505$, $AR_{\text{large}} = 0.708$), with noticeably lower AP on small objects ($AP_{\text{small}} = 0.375$). This pattern suggests that the anchor-free paradigm can match or exceed anchor-based models on irregular, well-resolved lesions, but small early-stage symptoms remain a primary source of error.

Per-class AP/AR further highlights the impact of the long-tailed distribution. Classes with abundant, visually distinctive samples such as *Cherry leaf*, *Blueberry leaf*, and *Raspberry leaf* reach AP scores above 0.80, whereas rare or visually subtle conditions such as *Soyabean leaf* and *Squash Powdery mildew leaf* fall below 0.20 AP. For some extremely underrepresented categories (e.g., *Tomato leaf*, *Tomato mold leaf*), AP is effectively undefined (NaN) on the validation split, confirming that data scarcity is a major limitation rather than purely an architectural issue.

4.2 Effect of CBAM on the Lightweight YOLOv5s

Table III reports the effect of inserting CBAM into the YOLOv5s backbone. Contrary to the initial hypothesis, CBAM degrades overall detection performance in this setting. Precision drops from 0.535 to 0.475 and recall from 0.676 to 0.635. The main metrics also decrease, with mAP@0.5 falling from 0.633 to 0.590 and mAP@0.5:0.95 from 0.485 to 0.434.

These results indicate that, under the limited-data and strong-augmentation regime considered here, the additional attention layers do not translate into more robust feature focusing. A plausible explanation is that CBAM introduces extra parameters and inductive bias that overfit to spurious patterns in the small training set, partially disrupting the pretrained COCO features that YOLOv5s relies on. In other words, attention does increase representational flexibility, but the available data may be insufficient to exploit it effectively on PlantDoc.

4.3 Optimization Strategies: Soft-NMS and Focal Loss on YOLOv5m

Table 4 analyzes the impact of applying Soft-NMS and focal loss to the higher-capacity YOLOv5m baseline. Replacing traditional NMS with Soft-NMS leads to a noticeable increase in precision (from 0.597 to 0.620) but a decrease in recall (from 0.667 to 0.624). The net effect on mAP is small: mAP@0.5 decreases slightly (0.647 to 0.641), whereas mAP@0.5:0.95 increases marginally (0.506 to 0.509). This behavior is consistent with the intended effect of Soft-NMS: it reduces duplicate and low-quality detections, yielding cleaner predictions at the cost of missing some overlapping lesions.

Introducing focal loss instead of the standard BCE also reshapes the precision–recall trade-off. Compared to the baseline, focal loss slightly lowers precision (0.597 to 0.585) and recall (0.667 to 0.650), but yields a small gain in mAP@0.5 (from 0.647 to 0.656) and in mAP@0.5:0.95 (from 0.506 to 0.510). These improvements are numerically modest, but they indicate that re-weighting hard examples can partially compensate for the long-tailed class distribution without fundamentally changing the architecture.

Overall, both Soft-NMS and focal loss act as fine-grained optimization tools that adjust where the model lies on the precision–recall spectrum rather than providing a dramatic boost in average precision. This supports our broader conclusion that optimization strategies alone cannot fully overcome the data scarcity and class imbalance in PlantDoc, and that architectural choices (e.g., anchor-free vs. anchor-based) and dataset design remain equally critical.

Table 1: Per-class performance for a subset of frequent and rare classes on the PlantDoc validation set (YOLOX-s).

Class	AP@0.5:0.95	AR@0.5:0.95
Cherry leaf	0.86643	0.90000
Blueberry leaf	0.83184	0.86333
Raspberry leaf	0.81303	0.84667
Soyabean leaf	0.13683	0.55000
Squash Powdery mildew leaf	0.17733	0.38421
Tomato leaf	–	–

Table 2: Overall detector performance on the PlantDoc validation set.

Model	mAP@0.5	mAP@0.5:0.95
YOLOv5s (anchor-based)	0.633	0.485
YOLOv5m (anchor-based)	0.647	0.506
YOLOX-s (anchor-free)	0.661	0.504

Table 3: Effect of CBAM on YOLOv5s on the PlantDoc validation set.

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv5s baseline	0.535	0.676	0.633	0.485
YOLOv5s + CBAM	0.475	0.635	0.590	0.434
Δ (CBAM–Base)	−0.060	−0.041	−0.043	−0.051

Table 4: Effect of Soft-NMS and focal loss on YOLOv5m on the PlantDoc validation set.

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv5m baseline (BCE + NMS)	0.597	0.667	0.647	0.506
YOLOv5m + Soft-NMS	0.620	0.624	0.641	0.509
YOLOv5m + Focal loss	0.585	0.650	0.656	0.510

5 Discussion

Overall, the three YOLO-style detectors achieve reasonable performance on the PlantDoc validation split despite the small dataset size and strong class imbalance. Scaling the anchor-based baseline from YOLOv5s to YOLOv5m consistently improves both mAP@0.5 and mAP@0.5:0.95, indicating that additional capacity is beneficial under these challenging conditions. The anchor-free YOLOX-s slightly surpasses YOLOv5m in mAP@0.5 while reaching comparable mAP@0.5:0.95, suggesting that a more flexible, anchor-free localization mechanism can match or exceed anchor-based detectors on irregular leaf lesions. Long-tail optimization strategies provide smaller but still meaningful gains: on YOLOv5m, replacing BCE with focal loss and substituting hard NMS with Soft-NMS both lead to modest improvements in COCO-style mAP@0.5:0.95 and allow finer control over the precision-recall trade-off. Per-class AP/AR analysis further shows that frequent, visually distinctive categories (e.g., *Cherry leaf*, *Blueberry leaf*, *Raspberry leaf*) reach high AP values above 0.8, whereas rare or subtle diseases such as *Soyabean leaf* and *Squash Powdery mildew leaf* remain below 0.2 AP, highlighting the impact of the long-tailed distribution.

At the same time, several interventions do not work as initially expected, and important limitations remain. Adding CBAM to the lightweight YOLOv5s backbone degrades precision, recall and both mAP metrics, likely because the extra attention layers increase model complexity without sufficient data to constrain them, partially disrupting useful COCO-pretrained features. Neither anchor-free detection nor focal loss and Soft-NMS fundamentally solve the small-object problem or the extreme class imbalance: AP for small lesions is still much lower than for large ones, and some extremely underrepresented classes (e.g., *Tomato leaf*, *Tomato mold leaf*) cannot be reliably evaluated at all. Methodologically, the study is also limited by the small dataset size, the highly long-tailed label distribution, and the absence of a separate test set; the reported numbers may therefore slightly overestimate true generalization, although relative comparisons remain meaningful because all models share the same held-out validation split. Future work could focus on data-centric strategies (e.g., targeted collection for rare diseases, class rebalancing or merging) and on specialized small-object modules or multi-scale feature fusion tailored to early-stage lesions, ideally on a larger, more balanced benchmark with standardized anchor-free annotations.

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